

Question	Marking guidance	Mark	Comments
09.1	Acceleration increases with time or distance ✓	1	
09.2	Uses $a = \frac{F}{m}$ ✓ 0.143 (m s ⁻²) ✓	2	Substitution into the equation gets the mark – condone power of ten
09.3	Work done is area under graph – stated or used ✓ Evidence of correct calculation eg $\frac{(0.2 + 4.4) \times 10^3}{2} .90$ ✓ 2.07×10^5 ✓	3	Alternative method: Work done = average force x distance Condone powers of ten error in 2nd mark. Must have convincing manipulation of powers of ten and must be at least 2 sf in 3rd mark
09.4	Use of $E_k = \frac{1}{2}mv^2$ ✓ 17(.2) (m s ⁻¹) ecf from 09.3 ✓	2	Alternative method: calculates average acceleration and uses appropriate equation of motion. Acceptable answer 16.9 (m s ⁻¹) for candidates that use 2×10^5
09.5	Friction (or rolling friction)(between wheels and road) remains constant ✓ Air resistance increases (with time or distance travelled) ✓	2	Just mentioning air resistance and friction without correctly describing the expected changes (or lack of change) gets 1 mark Accept (overall) resistive forces increase for 1 mark
Total		10	

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09.1	Work done is force $\times$ distance ✓ travelled in the direction of the force ✓	2	Accept $F \times s$ with symbols defined for MP1 Accept $F \times s \times \cos \theta$ with symbols defined for both marks
09.2	use of $(m)gh = \frac{1}{2} (m)v^2$ ($9.81 \times 5.5 = 0.5v^2$) OR use of $mg \sin \theta = ma$ with $v^2 = u^2 + 2as$ where a = acceleration down slope and $s = 17.5$ m ✓ $v = 10.4 \text{ m s}^{-1}$ ✓	2	Do not accept use of $v^2 = u^2 + 2as$ where $g = 9.81$ and $s = 5.5$ m answer to at least 3sf for second mark

09.3	Finds one energy correctly ✓ attempts to find difference in energies ✓ uses friction force = work done/distance ✓ 160 OR 162 (N) cao ✓	4	E.g. Predicted kinetic energy = $0.5mv_{\text{predicted}}^2$ = $0.5(75)(10.4^2)$ (= 4056 J) True kinetic energy = $0.5mv_{\text{true}}^2$ $0.5(75)(5.8^2)$ (= 1261 J) Friction force = $\frac{(4056-1261)}{17.2}$ Alternative for first mark: $mgh = 75(9.81)(5.5) = 4047 \text{ J}$ Work done = 2795 (J) scores MP1 and MP2
Total		8	

Question	Marking guidance	Mark	Comments
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15	C
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21	B
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22	C
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	mention of an appropriate circumstance (eg circular motion)✓		
02	$\text{kg m}^2 \text{s}^{-2}$ ✓ work = force × distance or other valid energy equation or a joule is a N m or ✓	2	Correct answer with no working scores 2 marks. Accept symbol equation using normal symbols without definition of symbols=

13	A
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14	D
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20	C
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Question	Marking guidance	Mark	Comments
06.1	Kinetic energy at start of the curved section is 1125 J OR Change in gravitational energy on the curved section is 863.3 J ✓ Adding their KE and GPE ✓ 1988.3 J ✓	3	Accept 2 or more sf

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06.2	Use of distance = work / force (= 2000 / 250) ✓ Distance = 8.00 m ✓	2	Allow candidate's value from 06.1 instead of 2000 J

Question	Marking guidance	Mark	Comments
06.3	(Kinetic energy of adult at the barrier = 460.8 J) Kinetic energy of adult at start of flat (= 460.8 × 10) = 4608 J ✓ Work done to friction is 4608 – 460.8 = 4147 J ✓ Friction (= 4147 / 12.5) = 332 N ✓ OR 9/4 × KE of child (2000) = 4500 J ✓ 0.1 × 4500 J = 450 J ✓ Friction = (4500 – 450) / 12.5 = 324 N ✓	3	If no marks are awarded, 1 mark can be awarded for calculating kinetic energy of adult at the barrier = 460.8 J Accept use of candidate's value of KE for child from 06.1

Question	Marking guidance	Mark	Comments
06.4	The change in momentum/impulse will be the same ✓ The time (over which the person stops) will be greater ✓ The rate of change of momentum will be less ✓ Force is proportional to the rate of change of momentum so the force will be less ✓ OR Work will be the same ✓ Force will be applied over a greater distance ✓ Force = work / greater distance ✓ constant work leads to lower force ✓	4	N2 approach: velocity at barrier will be the same ✓ The time (over which the person stops) will be greater ✓ so acceleration is less ✓ $F=ma$ so force is less. ✓ For the alternative scheme, do not credit statement of work = force $\times$ distance for third mark
Total		12	

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11	C	$\text{kg m}^2 \text{s}^{-2}$
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12	B	kinetic energy displacement
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18	D	20 <i>P</i>

20	D	$12t.$	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	use of $KE = \frac{1}{2}mv^2$ to determine v OR $27 \text{ (m s}^{-1}\text{)}$ seen ✓ use of $P = Fv$ with their v to determine F ✓ $1.4 \times 10^5 \text{ (N)}$ ✓	accept POT errors for MP1 and MP2 calculator value: $1.3663155499... \times 10^5 \text{ (N)}$; penalise rounding errors in final answer	3	AO2 × 3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	use of $\sin \theta = 1.5/100$ or angle calculated correctly = 0.86° ✓ use of acceleration = $g \sin \theta$ (for their θ) OR use of $F = ma$ with a force substituted ✓ $(-)0.15 \text{ (m s}^{-2}\text{)}$ ✓	accept the angle to the vertical used consistently throughout accept alternative MP1: use of $mgh = F_{net}d$ condone incorrect sign	3	AO2 × 3
Total			6	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
02	use of efficiency in a correct equation OR 5.25 (W) OR 10.5 (W) seen ✓ 16% or 0.16 ✓	eg $P = 7 \text{ W} \times 0.75$ Allow POT errors for MP1 Condone the factor of two incorrectly placed in MP1. Calculator value is 16.1538462	2	AO1 × 2
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	AO
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11	A	$m(g s - \frac{v^2}{2})$	AO3
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19	C	24 m	AO1
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20	D	$\text{kg m}^2 \text{s}^{-3}$	AO3
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21	C	4500 N	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
02	Attempts to calculate one area ✓ Correctly calculates both areas ✓ • above displacement axis: 3.53 to 3.61 (J) • below displacement axis: 0.68 to 0.70 (J) Finds difference between their areas ✓	Must be calculated to the displacement axis Correct answer that rounds to 2.8 to 2.9 (J)	3	2 × AO1 1 × AO3
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	Any two from ✓✓ - Uses $\Delta h = 8.2 \sin 15$ OR component of weight parallel to belt = $mg \sin 15$ correctly - Uses $\Delta E_p = mg\Delta h$ correctly for their calculated Δh OR Uses $\Delta E_p = Fs$ correctly for their calculated their component of weight parallel to belt - Divides their value of ΔE_p by 43 11 (10.7) (s) ✓		3	1 × AO1 1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	Any one from ✓ - Use output power = $\frac{31}{100} \times 1200$ - Divides their output power by 43 8 bags✓	Expect to see 372 W Condone POT error in MP1 Do not accept 9.	2	1 × AO1 1 × AO2
Total			5	

Question	Answers	Additional comments/Guidelines	Mark	AO
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14	B	The total increase in kinetic energy of the rocket is $F\alpha$	AO3
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16	B	2.7 MJ	AO2

24	B	$\frac{mgv}{\eta}$	AO3
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Total 14 marks

Question	Answers	Additional comments/Guidelines	Mark	AO
01	Any valid equation for power with valid SI units for at least one of the terms ✓ $\text{kg m}^2 \text{s}^{-3}$ ✓	$P = \frac{\text{Any energy equation}}{\text{time}}$ OR $P = \text{any force} \times \text{equation } v$ Do not penalise incorrect case of letters for mp1 All letters should be clearly lower case for mp2	2	AO1
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	AO
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20	C	15 kW	AO3
			AO2

22	B	77 W	AO2
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Question	Marking guidance	Mark	Comments
02.1	Use of power = Fv : $35000 = F1500/28$ ✓ 653(650) (N) ✓	2	Condone error in power of 10 in substitution
02.2	Streamline the structure of the car/design a new engine with greater power output ✓	1	

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